Interpretable Low-Rank Sparse Unmixing and Spatial Attention-Enhanced Difference Mapping Network for Hyperspectral Change Detection

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***Abstract*—Hyperspectral change detection is a key technique for recognizing differences in hyperspectral images (HSIs) of the same location over two or more periods of time, widely used in the fields of environmental monitoring, resource management, and urban planning. Although deep learning methods have shown excellent performance in such tasks, their intrinsic "black-box" nature, that is, their lack of understanding of the intrinsic physical processes of the surface system and the need for a large amount of training data to achieve good performance, limits their generalization to application scenarios that require a high degree of transparency and interpretability. To address this problem, we propose an Interpretable low-rank sparse unmixing and spatial attention-enhanced difference mapping (LSU-SADM) network. This framework effectively combines traditional optimization algorithms with deep learning mechanisms, bridging the gap between interpretable modeling and powerful data processing. Specifically, LSU-SADM incorporates two key modules. The first module, the low-rank sparse unmixing module (LSUM), which is unfolded from the low-rank sparse coding unmixing (LSCU) model, decomposes hyperspectral data into abundance matrices, optimized via the inexact augmented Lagrange multiplier (IALM) method. The two abundance matrices obtained from the LSUM are concatenated as input to the second module, the spatial attention-enhanced difference mapping module (SADMM). The SADMM, building on the principles of particle swarm optimization (PSO), integrates both global and local spatial information, refining the accuracy of the difference map for precise change detection. The final enhanced feature map is then fed into a classifier for detection results. Experimental results show that LSU-SADM outperforms current state-of-the-art methods. Codes are available at https://github.com/QingranCai/LSU-SADM**

***Index Terms*—Change detection, hyperspectral images (HSIs), hyperspectral unmixing, interpretable.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide rich spectral data across hundreds of bands, enabling precise detection of subtle changes in materials and surfaces. Hyperspectral change detection plays a vital role in monitoring environmental changes over time and is widely used in fields such as land-cover analysis [1], agricultural crop monitoring [2], and urban development [3].

Hyperspectral change detection methods are typically categorized into traditional and deep learning-based approaches. Traditional methods rely on statistical and mathematical models for pixel-wise comparisons or feature extraction to detect changes between bi-temporal images. While these methods are computationally efficient and interpretable, they often struggle with complex, nonlinear changes [4], [5], noise, and data variability, limiting their performance in dynamic or intricate scenarios.

Deep learning methods, especially convolutional neural networks (CNNs), have become popular for hyperspectral change detection due to their ability to automatically learn complex features from large datasets, excelling at detecting subtle changes. However, their "black-box" nature reduces interpretability, limiting trust and application in domains where transparency is essential [6].

To address the aforementioned challenges, we introduce a novel framework called the Interpretable low-rank sparse unmixing and spatial attention-enhanced difference mapping (LSU-SADM) network. This framework harmonizes the interpretability of traditional methods with the high-dimensional data processing capabilities of deep learning techniques, targeting effective hyperspectral change detection. Specifically, our approach initiates with a hyperspectral low-rank sparse coding unmixing (LSCU) model, employing the inexact augmented Lagrange multiplier (IALM) [7] method to unfold the LSCU model into a low-rank sparse unmixing module (LSUM). This module maps dual-temporal phase images into abundance matrices and initially generates a feature map as input to the next module via a concatenation operation. Subsequently, we develop the spatial attention-enhanced difference mapping module (SADMM), which thoroughly incorporates both global and local information to refine hyperspectral change detection. This module iteratively produces a weighted final feature map, enhancing the generalization ability of the

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model. The principal contributions of this article are outlined as follows.

1) We propose an LSU-SADM, a novel framework that integrates traditional optimization with deep learning. It features two interpretable modules: one for hyperspectral unmixing to extract abundance matrices, and another for spatial attention to enhance spatial correlations. This dual-module design improves accuracy and provides a deeper understanding of surface system changes, combining high performance with interpretability.
2) A hyperspectral LSCU model is designed to handle complex spectral data, implemented as the LSUM using the IALM method. The LSUM enhances the decomposition of material components, delivering richer and more precise feature information critical for change detection, while improving spectral data utilization and maintaining result clarity.
3) The SADMM is designed to refine change detection by integrating local and global spatial information. Using particle swarm optimization (PSO), the SADMM adjusts weight distribution dynamically, improving the sensitivity of the model to subtle and significant changes while enhancing reliability and providing clearer insights into the detection process.

The rest of this article is organized as follows. The related works are briefly reviewed in Section II. Section III illustrates the proposed LSU-SADM in detail. Section IV introduces the dataset and performs a comparative and ablation study. Section V gives the conclusion.

II. Related Work

A. Traditional Methods

In the field of hyperspectral change detection, traditional methods can be categorized into four major types: algebra-based, transformation-based, classification-based, and deep learning-based approaches.

Algebra-based methods detect changes by directly comparing pixel values over time, typically through image differencing or ratioing. A representative method is change vector analysis (CVA) [8], which computes spectral differences in a multidimensional space. However, these methods are sensitive to image misregistration and require careful threshold selection.

Transformation-based methods map the data to new domains to enhance the separation of changed and unchanged areas. For example, PCA [9] reduces dimensionality to extract dominant features, while SFA [10] targets slowly varying signals, useful for long-term change detection. IR-MAD [11], built upon CCA [12], iteratively adjusts weights to reduce noise and false positives, but at a higher computational cost.

Classification-based methods view change detection as a supervised task and apply classifiers such as the SVM [13] and MLC [14]. Though effective, their performance hinges on training data quality and representativeness and can be affected by preprocessing steps and class imbalance.

B. Deep Learning-Based Methods

Although the HSI provides detailed spectral information, its high dimensionality and spectral interdependence pose challenges for change detection [15]. Traditional methods rely on manual feature extraction and substantial computational resources, which are impractical for real-world applications. Recent advances in deep learning, particularly CNNs and autoencoders, offer efficient and accurate solutions for hyperspectral change detection. These models automatically learn complex spatial and spectral features, enabling finer classification and more sensitive change detection [16], [17], [18]. Wang et al. [19] introduced the SSA-SiamNet framework, which combines spectral–spatial features and attention modules to minimize redundancy. Lin et al. [20] developed bilinear CNNs (BCNNs) that integrate bi-temporal hyperspectral data using linear operations and neighborhood context. Huang et al. [21] proposed the SEIFNet, a Siamese network model utilizing an encoder–decoder structure to accentuate temporal differences and combine multiscale features. Similarly, Qu et al. [22] designed ML-EDAN, which integrates LSTM networks to capture spatiotemporal spectral features and temporal correlations. Yang et al. [23] introduced DMPs2raN, optimizing network performance by fusing multiscale and multilevel features for detailed spatial and semantic understanding. Wang et al. [24] proposed CSDBF, a two-branch framework combining pixel and super-pixel features for detecting both subtle and large-scale changes. Qu et al. [25] developed CMJAN, an unsupervised domain-adaptive model that reduces domain shift using cycle-refined pseudo-labels. Luo et al. [26] presented the DCENet, a semi-supervised approach enhancing feature contrast with differential fusion attention and Siamese modules. Wu et al. [27] proposed DIEFEN, leveraging differential multihead attention for improved feature exchange and change detection. Yao et al. [28] introduced ECDBS, integrating band selection and spatial attention for more efficient and accurate hyperspectral change detection.

The high dimensionality and large data volume of HSIs pose significant processing challenges for change detection [29]. Hyperspectral unmixing is one of the key techniques to deal with this challenge. It aims to identify and quantify the material components contained in each pixel from hyperspectral data. Conventional hyperspectral unmixing methods, such as linear and nonlinear unmixing [30], although effective in many cases, may not be sensitive or accurate enough when dealing with complex scenes or dynamic changes [31]. Tao et al. [32] proposed A2SAN, a hyperspectral unmixing network based on spectral–spatial attention, which adaptively captures contextual information without relying on predefined endmembers, leading to improved unmixing performance. Yang and Zhang [33] introduced a dual embedding transformer unmixing network that models spectral–spatial correlations more effectively by applying cross-attention between endmember and abundance embeddings, achieving better results on complex scenes. Hu et al. [34] proposed an excellent unsupervised multitask learning network named BCG-Net, which integrates temporal correlation into spectral unmixing for the first time and achieves good results in binary and multiclass

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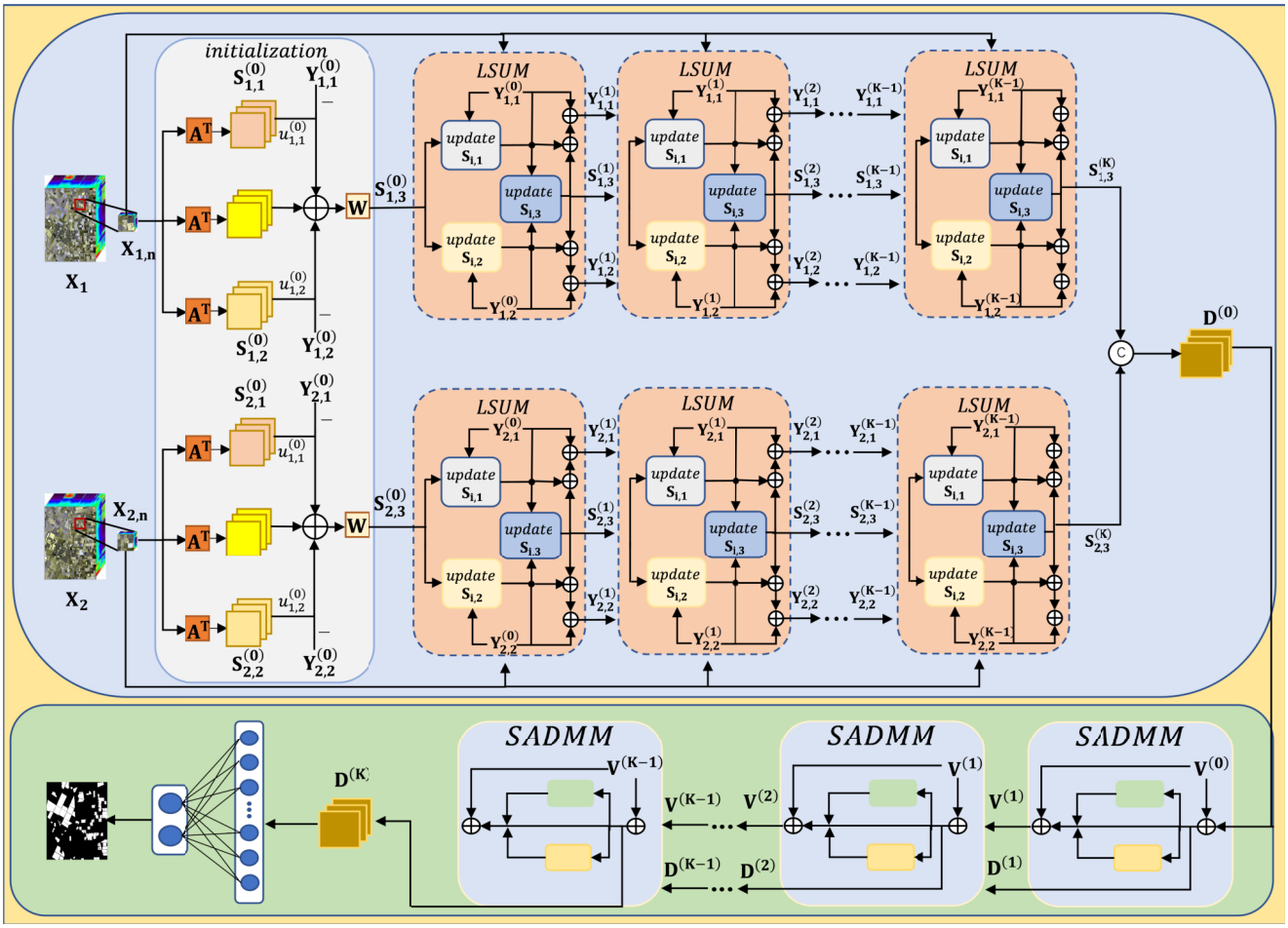


Fig. 1. Architecture of LSU-SADM for hyperspectral unmixing change detection, consisting of the LSUM and SADMM. The LSUM processes bi-temporal HSIs to generate abundance matrices with low-rank sparse properties, making the unmixing process transparent. The SADMM performs change detection on the abundance matrices at the subpixel level by incorporating spatial attention mechanisms and the traditional algorithm, ensuring both accuracy and interpretability in the final detection.

change detection. Dong et al. [35] proposed an AMCAN-HMSchA for subpixel-level change detection by mapping temporal differences to abundance matrices and adaptively reweighting image features to highlight changes and suppress unchanged areas.

Despite the progress of deep learning methods, they often function as black boxes, lacking transparency in their decision-making process, which restricts their trustworthiness in critical applications such as environmental monitoring, where interpretability is essential. Moreover, the complexity of these models complicates their design, as configuring network structures and hyperparameters typically requires extensive trial and error, leading to high computational costs and challenging optimization. Their strong reliance on specific data distributions also limits generalizability across different datasets, further underscoring the need for more interpretable and adaptable approaches.

C. Model-Based Deep Learning Methods

Recently, interpretable methods have shown their value in hyperspectral data processing. In unmixing, MNN-BU [36] unfolded the ISTA optimization of the linear mixing model (LMM) for sparse abundance estimation, while the SNMF-Net [37] extended this idea by unfolding Lp-NMF with alternating endmember and abundance updates. Both frameworks offer clear physical interpretability but lack low-rank modeling and do not explicitly consider spatial information, which is crucial for hyperspectral change detection, where spatial context greatly influences the accuracy and reliability of results. In change detection, the DLDINet [38] employed a dictionary learning-guided deep network to enhance interpretability in abundance estimation but did not explicitly model spatial information. The MSCSCNet [39] combined multiscale sparse coding with a teacher–student strategy to leverage both labeled and unlabeled data. However, the interpretability of both methods is mainly confined to the abundance extraction stage, lacking clear physical explanations in the detection phase. This limits their ability to intuitively explain the causes and contributions of detected changes, thereby reducing transparency and reliability.

III. Proposed Method

A. Overview

Hyperspectral change detection aims to identify variations between bi-temporal HSIs, but traditional methods often require manual parameter tuning and may struggle

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with noise sensitivity, while deep learning approaches are typically black-box models with limited interpretability. To address these challenges, we propose LSU-SADM, a two-stage framework combining model-driven unmixing and spatial attention-enhanced difference mapping, as shown in Fig. 1. In the first stage, the LSUM processes each HSI independently to generate abundance matrices, ensuring accurate spectral decomposition through an interpretable unmixing process. In the second stage, the concatenated abundance matrices from the first stage are input into the SADMM, which refines the change detection process at the subpixel level. This process remains interpretable as the SADMM leverages spatial attention mechanisms to focus on significant spatial features and incorporates traditional optimization algorithms to enhance spatial coherence. By maintaining transparency in the weighting and optimization of spatial features, the SADMM ensures that the change detection results are both accurate and interpretable. Finally, the processed feature map is passed through a fully connected layer to generate the final change detection map. This structured approach effectively balances spectral unmixing precision, spatial feature extraction, and interpretability, providing a more transparent and reliable solution for hyperspectral change detection.

B. Problem Formulation

Spectral mixing [40], where multiple materials within a pixel contribute to the observed spectrum, is a common challenge in change detection, often complicating the accurate identification of changes [41]. This phenomenon can obscure subtle changes by blending spectral signatures, making it difficult to accurately identify individual materials. Hyperspectral unmixing addresses this issue by separating the mixed spectral data into constituent materials (endmembers) and determining their abundance in each pixel. This process is essential for improving the precision of change detection in complex scenes, as it helps to clarify material variations over time. The most commonly used model for hyperspectral unmixing is the LMM [42], which assumes that the observed spectrum at each pixel is a linear combination of pure spectral signatures from different materials. The fidelity of this model is crucial, as accurately quantifying material changes can be directly linked to actual physical alterations in the observed area. The LMM can be mathematically expressed as

$$\begin{cases} \mathbf{X} = \mathbf{AS} + \mathbf{E} \\ \text{s.t. } \mathbf{A} \geq 0, \mathbf{S} \geq 0, \mathbf{1}_P^T \mathbf{S} = \mathbf{1}_N^T \end{cases} \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{B \times N}$ represents the observed data matrix with B spectral bands and N pixels, $\mathbf{A} \in \mathbb{R}^{B \times P}$ is the matrix containing the spectral signatures of the P endmembers, $\mathbf{S} \in \mathbb{R}^{P \times N}$ denotes the abundance matrix, where each column represents the proportion of endmembers in a corresponding pixel, and $\mathbf{E} \in \mathbb{R}^{B \times N}$ is the error or noise matrix accounting for model inaccuracies and observational noise. $\mathbf{A} \geq 0$ and $\mathbf{S} \geq 0$ imply that each of our elements must be nonnegative, which reflects the fact that negative ratios are physically meaningless. $\mathbf{1}_P^T \mathbf{S} = \mathbf{1}_N^T$ indicates that the sum of the abundance fractions for each pixel must equal 1 to ensure that the combined proportion of all inclusions accounts for the full composition of the pixel.

The LMM was chosen for its simplicity, interpretability, and computational efficiency [43], [44], [45], [46]. It assumes that each pixel's spectrum is a linear combination of pure material signatures, which results in an optimization that is computationally simpler compared to nonlinear models with more complex optimization procedures. The gradients of the LMM with respect to its parameters involve matrix multiplications, making it highly efficient and compatible with the operations commonly used in deep learning models, such as convolutional layers. This compatibility not only enhances efficiency but also ensures interpretability, as the LMM's abundance matrix directly represents material proportions in each pixel, which is crucial for change detection tasks. Despite possible nonlinear effects in real-world data, the LMM remains robust and optimally suited for our framework, enabling efficient and interpretable modeling of spectral mixing.

Based on the LMM, we propose a hyperspectral LSCU model that efficiently combines low-rank and sparse matrix decompositions to enhance the accuracy and computational efficiency of hyperspectral unmixing. The objective function can be formulated as

$$\begin{cases} \min\limits_{\mathbf{S}_{1-3}} \dfrac{1}{2} \|\mathbf{X} - \mathbf{AS}_3\|_F^2 + \lambda_1 \|\mathbf{S}_1\|_* + \lambda_2 \|\mathbf{S}_2\|_{1,1} \\ \text{s.t.} \quad \mathbf{S}_3 = \mathbf{S}_1, \quad \mathbf{S}_3 = \mathbf{S}_2 \end{cases} \tag{2}$$

where $\|\mathbf{S}_1\|_*$ is a regularization of the nuclear norm of $\mathbf{S}_1$, $\|\mathbf{S}_2\|_{1,1}$ imposes sparsity on $\mathbf{S}_2$ by penalizing the $L_{1,1}$ norm, and λ_1 and λ_2 are the regularization parameters. The introduction of $\mathbf{S}_1$, $\mathbf{S}_2$, and $\mathbf{S}_3$ addresses conflicts between low-rank and sparse regularizers, which, when applied to the same variable, create a nonsmooth, nonconvex optimization problem. By decoupling these constraints, each regularizer independently achieves its objective, while equality constraints ensure consistency. This simplifies optimization, improves stability, and ensures robust, efficient solutions.

The LSCU model unfolds into the LSUM using the IALM method. The LSUM independently unmixes bi-temporal HSIs, preserving the spectral characteristics of each time point without temporal constraints. This independent processing reduces spectral mixing and improves abundance extraction, particularly in complex scenes with subtle temporal changes. The dual low-rank and sparse decomposition enhances the model's ability to capture key spectral patterns and dominant materials, ensuring accurate change detection without interference from temporal dependencies.

By focusing on material abundance changes, this formulation refines the detection of subtle variations, directly linking them to the change detection task. The abundance matrices from the LSUM are concatenated and fed into the SADMM, which uses PSO to integrate spatial information, further enhancing detection accuracy while maintaining interpretability.

C. Optimization for the LSCU Model

To solve the optimization problem in (2), we use the traditional IALM method, and the augmented Lagrangian function

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is defined as

$$L(\mathbf{S}_{1-3}) = \frac{1}{2}\|\mathbf{X}-\mathbf{A}\mathbf{S}_3\|_F^2 + \lambda_1\|\mathbf{S}_1\|_* + \lambda_2\|\mathbf{S}_2\|_{1,1} + \operatorname{tr}\left(\mathbf{Y}_1^T(\mathbf{S}_3-\mathbf{S}_1)\right) + \frac{u_1}{2}\|\mathbf{S}_3-\mathbf{S}_1\|_F^2 + \operatorname{tr}\left(\mathbf{Y}_2^T(\mathbf{S}_3-\mathbf{S}_2)\right) + \frac{u_2}{2}\|\mathbf{S}_3-\mathbf{S}_2\|_F^2 \quad (3)$$

where $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are Lagrange multipliers and $u_1 > 0$ and $u_2 > 0$ are two penalty parameters.

1) *Update* $\mathbf{S}_1$*:* To obtain $\mathbf{S}_1^{(k+1)}$, we can solve the following subproblem:

$$\mathbf{S}_1^{(k+1)} = \arg\min_{\mathbf{S}_1}\left(\frac{\lambda_1}{u_1}\left\|\mathbf{S}_1^{(k)}\right\|_* + \frac{1}{2}\left\|\mathbf{S}_1^{(k)} - \left(\mathbf{S}_3^{(k)} + \frac{1}{u_1}\mathbf{Y}_1^{(k)}\right)\right\|_F^2\right). \quad (4)$$

To solve the above optimization subproblem, we employ the singular value soft thresholding (SVT) technique. Specifically, we first perform a singular value decomposition of the synthesis matrix

$$\mathbf{M} = \mathbf{S}_3^{(k)} + \frac{1}{u_1}\mathbf{Y}_1^{(k)} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (5)$$

where $\mathbf{\Sigma}$ can be expressed as

$$\mathbf{\Sigma} = \mathbf{U}^T\mathbf{M}\mathbf{V}. \quad (6)$$

Then, a singular value soft threshold is applied to $\mathbf{\Sigma}$, and for each singular value σ_i in $\mathbf{\Sigma}$, it is treated as follows:

$$\sigma_i^* = \max\left(\sigma_i - \frac{\lambda_1}{u_1}, 0\right). \quad (7)$$

Since it is a diagonal matrix, the updated $\mathbf{\Sigma}$ can be expressed as

$$\mathbf{\Sigma}^* = \operatorname{diag}\left(\max\left(\mathbf{\Sigma} - \frac{\lambda_1}{u_1}\mathbf{I}, 0\right)\right). \quad (8)$$

Therefore, we use $\mathbf{\Sigma}^*$ to reconstruct $\mathbf{S}_1$

$$\mathbf{S}_1^{(k+1)} = J_{\lambda_1/u_1}(\mathbf{M}) = \mathbf{U}\mathbf{\Sigma}^*\mathbf{V}^T \quad (9)$$

where $J_{\lambda_1/u_1}(\cdot)$ is the singular value soft threshold operation. After simplification, we can get the final mathematical expression for $\mathbf{S}_1^{(k+1)}$

$$\mathbf{S}_1^{(k+1)} = \mathbf{U}\operatorname{diag}\left(\max\left(\mathbf{U}^T\left(\mathbf{S}_3 + \frac{1}{u_1}\mathbf{Y}_1\right)\mathbf{V} - \frac{\lambda_1}{u_1}\mathbf{I}, 0\right)\right)\mathbf{V}^T \quad (10)$$

where $\mathbf{U}, \mathbf{U}^T, \mathbf{V}, \mathbf{V}^T \in \mathbb{R}^{P\times P}$.

2) *Update* $\mathbf{S}_2$*:* To obtain $\mathbf{S}_2^{(k+1)}$, we can solve the following subproblem:

$$\mathbf{S}_2^{(k+1)} = \arg\min_{\mathbf{S}_2}\left(\frac{\lambda_2}{u_2}\left\|\mathbf{S}_2^{(k)}\right\|_{1,1} + \frac{1}{2}\left\|\mathbf{S}_2^{(k)} - \left(\mathbf{S}_3^{(k)} + \frac{1}{u_2}\mathbf{Y}_2^{(k)}\right)\right\|_F^2\right). \quad (11)$$

To solve the above optimization subproblem, we use the soft thresholding algorithm. Specifically, we take a soft thresholding operation for each element of $(\mathbf{S}_3^{(k)} + (1/u_2)\mathbf{Y}_2^{(k)})$ to obtain $\mathbf{S}_2^{(k+1)}$

$$\mathbf{S}_2^{(k+1)} = \mathbf{S}_{\lambda_2/u_2}\left(\mathbf{S}_3^{(k)} + \frac{1}{u_2}\mathbf{Y}_2^{(k)}\right) \quad (12)$$

where $S_{\lambda_2/u_2}(\cdot)$ is the soft threshold operation. Equation (12) can be further expressed as

$$\mathbf{S}_2^{(k+1)} = \operatorname{sign}\left(\mathbf{S}_3^{(k)} + \frac{1}{u_2}\mathbf{Y}_2^{(k)}\right) \times \max\left(0, \left(\mathbf{S}_3^{(k)} + \frac{1}{u_2}\mathbf{Y}_2^{(k)}\right) - \frac{\lambda_2}{u_2}\mathbf{I}\right). \quad (13)$$

3) *Update* $\mathbf{S}_3$*:* To obtain $\mathbf{S}_3^{(k+1)}$, we can solve the following subproblem:

$$\mathbf{S}_3^{(k+1)} = \arg\min_{\mathbf{S}_3} f(\mathbf{S}_3) \quad (14)$$

where

$$f(\mathbf{S}_3) = \frac{1}{2}\left\|\mathbf{X}-\mathbf{A}\mathbf{S}_3^{(k+1)}\right\|_F^2 + \operatorname{tr}\left(Y_1^T\left(\mathbf{S}_3^{(k+1)} - \mathbf{S}_1^{(k+1)}\right)\right) + \frac{u_1}{2}\left\|\mathbf{S}_3^{(k+1)} - \mathbf{S}_1^{(k+1)}\right\|_F^2 + \operatorname{tr}\left(Y_2^T\left(\mathbf{S}_3^{(k+1)} - \mathbf{S}_2^{(k+1)}\right)\right) + \frac{u_2}{2}\left\|\mathbf{S}_3^{(k+1)} - \mathbf{S}_2^{(k+1)}\right\|_F^2. \quad (15)$$

To solve the above optimization subproblem, we use the gradient descent method

$$\nabla_{\mathbf{S}_3} f(\mathbf{S}_3) = \mathbf{A}^T\mathbf{A}\mathbf{S}_3^{(k+1)} - \mathbf{A}^T X + \mathbf{Y}_1^{(k)} + u_1\left(\mathbf{S}_3^{(k+1)} - \mathbf{S}_1^{(k+1)}\right) + \mathbf{Y}_2^{(k)} + u_2\left(\mathbf{S}_3^{(k+1)} - \mathbf{S}_2^{(k+1)}\right) = 0. \quad (16)$$

The equation can be organized as

$$\mathbf{W}^{-1}\mathbf{S}_3^{(k+1)} = \mathbf{A}^T\mathbf{X} - \mathbf{Y}_1^{(k)} - \mathbf{Y}_2^{(k)} + u_1\mathbf{S}_1^{(k+1)} + u_2\mathbf{S}_2^{(k+1)} \quad (17)$$

where $\mathbf{W}^{-1}$ can be expressed as

$$\mathbf{W}^{-1} = \mathbf{A}^T\mathbf{A} + u_1 I + u_2\ I. \quad (18)$$

Therefore, $\mathbf{S}_3^{(k+1)}$ can be expressed as

$$\mathbf{S}_3^{(k+1)} = \mathbf{W}\mathbf{G} \quad (19)$$

where

$$\mathbf{G} = \mathbf{A}^T\mathbf{X} - \mathbf{Y}_1^{(k)} - \mathbf{Y}_2^{(k)} + u_1\mathbf{S}_1^{(k+1)} + u_2\mathbf{S}_2^{(k+1)}. \quad (20)$$

4) *Update Multipliers:* According to the IALM method, the multipliers are updated as follows:

$$\begin{aligned}\mathbf{Y}_1^{(k+1)} &= \mathbf{Y}_1^{(k)} + u_1\left(\mathbf{S}_3^{(k+1)} - \mathbf{S}_1^{(k+1)}\right)\\ \mathbf{Y}_2^{(k+1)} &= \mathbf{Y}_2^{(k)} + u_2\left(\mathbf{S}_3^{(k+1)} - \mathbf{S}_2^{(k+1)}\right)\\ u_1 &= \rho u_1\\ u_2 &= \rho u_2\end{aligned} \quad (21)$$

where $\rho > 1$ is a constant factor used to progressively increase the penalty parameters u_1 and u_2, thereby enhancing the enforcement of the equality constraints across iterations.

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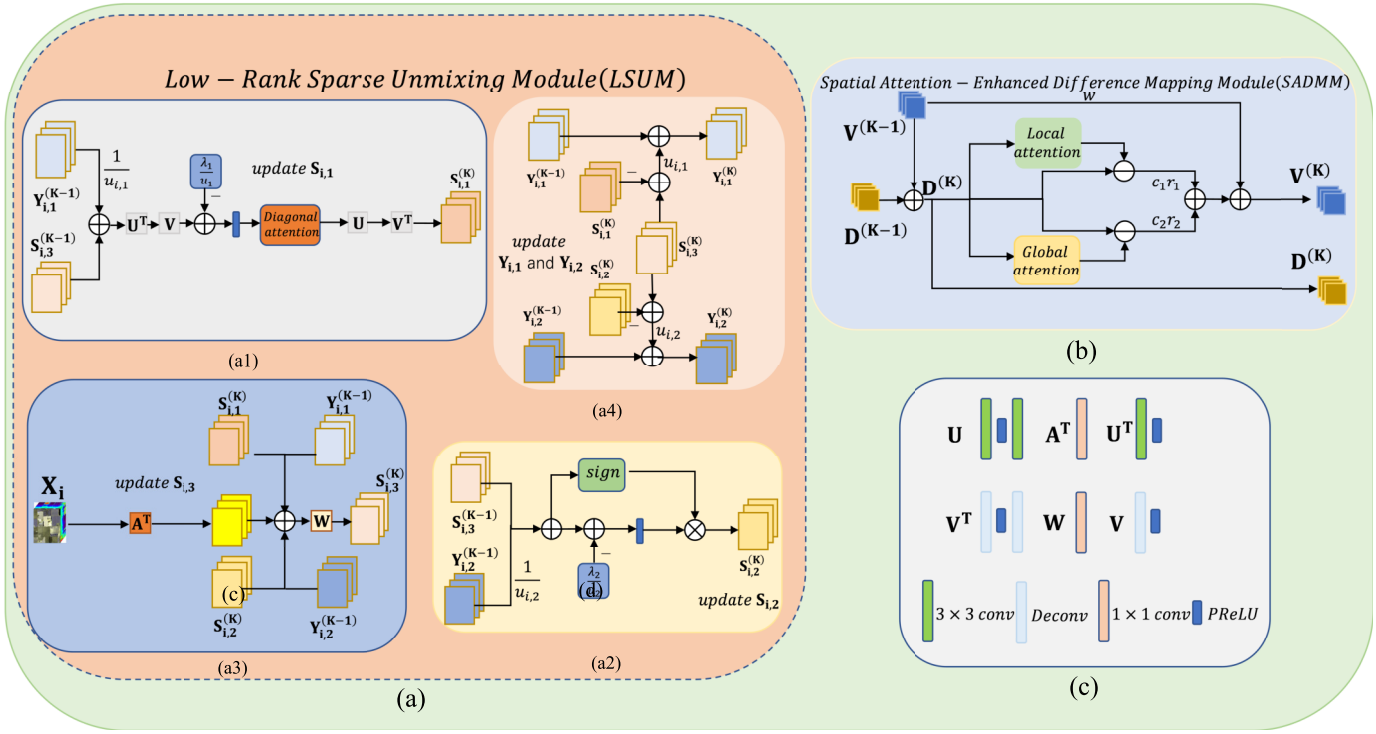


Fig. 2. (a) Architecture of the LSUM, which independently unmixes bi-temporal HSIs using low-rank sparse decomposition, ensuring interpretability and robust spectral separation. (b) Architecture of the SADMM, which refines extracted features using spatial attention and PSO to enhance change sensitivity and spatial consistency. (c) Explanation of key components, including convolutional and transformation operations used in the LSUM.

D. Modular Design of the LSUM

In this study, we emphasize the interpretability of the model by integrating traditional optimization methods with deep learning networks, ensuring that each module and operation unit is transparent and intuitive both in terms of technical efficiency and physical significance. Specifically, we utilize the powerful fitting capabilities of deep learning to refine the model parameters, ensuring that they accurately reflect real physical processes such as the distribution of major materials and the prominence of key substances. This approach not only enhances the interpretability and the transparency of the model in applications but also improves its practicality and scientific value by effectively balancing performance with transparency. Further details can be found in Fig. 2.

For (10), both matrix multiplication and convolution are essentially weighted sum operations, so we use multilayer convolution operations to simulate $\mathbf{U}$, $\mathbf{U}^T$, $\mathbf{V}$, and $\mathbf{V}^T$ [47], respectively, and $max(x, 0)$ can be expressed as the activation function of the nonlinear leakage rectified linear unit (ReLU). Additionally, since $diag(\cdot)$ serves as a diagonalization operator, we introduce the adaptive diagonal element emphasis module (ADEEM) to enhance diagonal elements using a diagonal bias mask while balancing their effect with a decay mask. This improves nonlinearity and feature extraction. The ADEEM first applies an attention mechanism to generate a base attention map, which adaptively highlights important features. The diagonal bias mask reinforces key diagonal elements, while the decay mask prevents overemphasis by controlling attenuation in nondiagonal regions. Importantly, the enhancement strength (strength) and decay factor (base decay) are learnable parameters, optimized during training for adaptive feature adjustment.

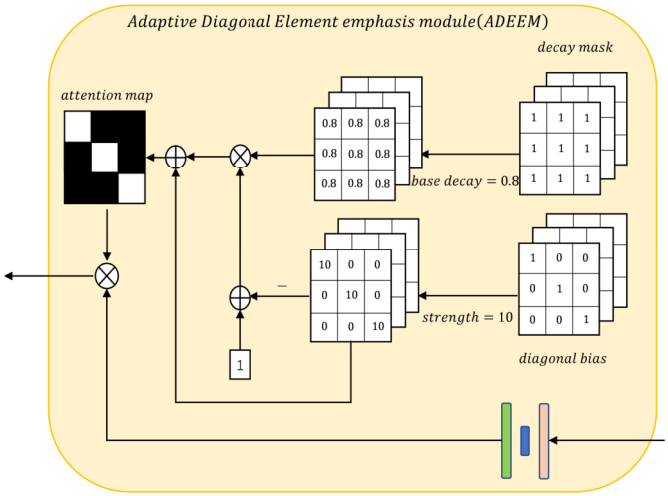


Fig. 3. Architecture of the ADEEM, which enhances diagonal elements in attention maps using a learnable strength parameter and a base decay factor. The diagonal bias emphasizes key elements, while the decay mask regulates surrounding values, refining feature representation and improving structured dependency capture.

This ensures stable optimization and enhances hyperspectral change detection accuracy. Specific details of the module can be seen in Fig. 3. The above process can be expressed as

$$\begin{aligned}\mathbf{E}_1 &= C_2\left(C_1\left(\mathbf{S}_3^{(k)} + \frac{1}{u_1}\mathbf{Y}_1\right)\right)\\ \mathbf{E}_2 &= \mathrm{ReLU}\left(\mathbf{E}_1 - \frac{\lambda_1}{\mathrm{u}_1}\mathbf{I}\right)\\ \mathbf{S}_1^{(k+1)} &= C_4\left(C_3\left(DA\left(\mathbf{E}_2\right)\right)\right)\end{aligned} \tag{22}$$

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where C_1, C_2, C_3, and C_4 represent convolutional layers with both input and output channels set to P. *DA* represents the ADEEM. The specific details of updating $\mathbf{S_1}$ can be seen in Fig. 2(a1).

For (13), we also use the ReLU activation function to represent $max(x, 0)$. Meanwhile, the sign function is meant to judge the positivity and negativity of the value of each element in the matrix, and multiply the before and after matrices to do the update of $\mathbf{S}_2$. The above process can be expressed as

$$\begin{aligned} \mathbf{F}_1 &= \mathbf{S}_3^{(k)} + \frac{1}{u_2}\mathbf{Y}_2^{(k)} \\ \mathbf{F}_2 &= \mathrm{ReLU}\left(\mathbf{F_2} - \frac{\lambda_1}{\mathbf{u_2}}\mathbf{I}\right) \\ \mathbf{F}_3 &= S\left(\mathbf{F}_1\right) \\ \mathbf{S}_2^{(k+1)} &= \mathbf{F}_3\mathbf{F}_2 \end{aligned} \tag{23}$$

where S denotes the absolute value of the elements in the matrix. The specific details of updating $\mathbf{S_2}$ can be seen in Fig. 2(a2).

For (19), we use convolutional layers to represent the matrices $\mathbf{W}$ and $\mathbf{A}^T$, respectively. The above process can be expressed as

$$\begin{aligned} \mathbf{G} &= C_5\left(\mathbf{X}\right) - \mathbf{Y}_1^{(k)} - \mathbf{Y}_2^{(k)} + u_1\mathbf{S}_1^{(k+1)} + u_2\mathbf{S}_2^{(k+1)} \\ \mathbf{S}_3^{(k+1)} &= C_6\left(\mathbf{G}\right) \end{aligned} \tag{24}$$

where C_5 represents a convolutional layer with an input channel of B and an output channel of P, and C_6 represents a convolutional layer with both input and output channels set to P. The specific details of updating $\mathbf{S_3}$ can be seen in Fig. 2(a3). Additionally, Fig. 2(a4) illustrates the updating process of (21), while Fig. 2(c) highlights the key components of the module. After K iterations, the abundance matrices obtained from the diachronic phase are concatenated to initially obtain the feature map. The algorithm flow of the LSUM is described in Algorithm 1, where λ_1, λ_2, ρ_1, and ρ_2 are learnable parameters that dynamically adjust during training to balance the effects of low-rank and sparse constraints. $u_{1,1}$, $u_{1,2}$, $u_{2,1}$, and $u_{2,2}$ are not fixed but iteratively updated throughout the optimization process.

Algorithm 1 Optimization Algorithm for Hyperspectral Unmixing Using Low-Rank Sparse Coding

Input: Bitemporal HSIs $\mathbf{X}_1$ and $\mathbf{X}_2$;
Output: The initial feature map $\mathbf{D}$;

Set regularization parameters $\lambda_1 > 0$, $\lambda_2 > 0$;
Set penalty parameters $u_{1,1} > 0$, $u_{1,2} > 0$, $u_{2,1} > 0$, $u_{2,2} > 0$;
Set augmented Lagrange multiplier update parameters $\rho_1 > 1$, $\rho_2 > 1$;
Initialize $\mathbf{S}_{1,1}^{(0)} = \mathbf{A}^T\mathbf{X}_1$, $\mathbf{S}_{1,2}^{(0)} = \mathbf{A}^T\mathbf{X}_1$;
Initialize $\mathbf{S}_{2,1}^{(0)} = \mathbf{A}^T\mathbf{X}_2$, $\mathbf{S}_{2,2}^{(0)} = \mathbf{A}^T\mathbf{X}_2$;
Initialize $\mathbf{Y}_{1,1}^{(0)} = 0$, $\mathbf{Y}_{1,2}^{(0)} = 0$, $\mathbf{Y}_{2,1}^{(0)} = 0$, $\mathbf{Y}_{2,2}^{(0)} = 0$;
Set iteration count $k = 0$;
while not converged **do**
  for $i = 1, 2$ **do**
    Update $\mathbf{S}_{i,1}$ via (10);
    Update $\mathbf{S}_{i,2}$ via (13);
    Update $\mathbf{S}_{i,3}$ via (19);
    Update multipliers $\mathbf{Y}_{i,1}$ and $\mathbf{Y}_{i,2}$ via (21);
  end for
  Update penalty parameters $u_{1,1}$, $u_{1,2}$, $u_{2,1}$, and $u_{2,2}$ via(21)
  $k = k + 1$;
end while
Compute the feature map $\mathbf{D}$ **via:**

$$\mathbf{D} = \mathrm{Concat}(\mathbf{S}_{1,3}, \mathbf{S}_{2,3})$$

return The initial feature map $\mathbf{D}$.

E. Spatial Attention-Enhanced Difference Mapping Module

The proposed LSUM effectively models the spectral mixing process with strong interpretability, as it is derived from a well-established optimization formulation. However, this module primarily focuses on spectral features and does not explicitly utilize spatial information, which is crucial in hyperspectral change detection due to the structured nature and contextual dependencies of changes across neighboring pixels. To address this limitation, we introduce the SADMM, as shown in Fig. 2(b). This module enhances sensitivity to spatially coherent changes by integrating both local and global attention mechanisms, while preserving the interpretability of the overall framework.

Specifically, the SADMM leverages deep learning networks to dynamically refine change detection strategies, enhancing the model's responsiveness to spatial changes in hyperspectral data. A critical element of the SADMM is the use of the traditional PSO algorithm, which has proved particularly effective for exploiting spatial information [48]. PSO's strengths lie in its ability to balance exploration and exploitation within a high-dimensional solution space. It simulates the social behavior of particles, where each particle adjusts its position based on both its own past experiences and the shared knowledge of the group. This mechanism enables PSO to search for both local and global solutions efficiently.

PSO's strength lies in its ability to dynamically adjust spatial attention weights, allowing the algorithm to focus on both local and global spatial relationships. This is crucial for hyperspectral change detection, where detecting both subtle material variations and broader land use shifts is essential. By integrating PSO into the SADMM, we optimize spatial information processing, ensuring accurate detection of significant changes. Moreover, PSO enhances model interpretability. Its transparent optimization of spatial weights provides clear, understandable results, which are crucial for applications requiring trust and clarity, such as environmental monitoring. The above process can be expressed as

$$\begin{aligned} x_i^{(t+1)} &= x_i^{(t)} + v_i^{(t+1)} \\ v_i^{(t+1)} &= wv_i^{(t)} + c_1r_1\left(p_i - x_i^{(t)}\right) + c_2r_2\left(g - x_i^{(t)}\right) \end{aligned} \tag{25}$$

where $x_i^{(t)}$ represents the position of particle i at iteration t, $v_i^{(t+1)}$ represents the velocity of particle i at iteration $t+1$, w is the inertia weight, which helps to control the impact of

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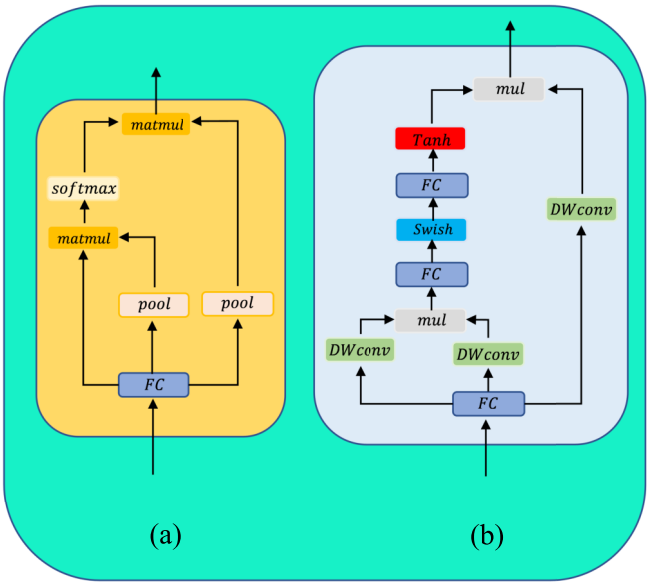


Fig. 4. (a) Global spatial attention module, which captures long-range dependencies by computing spatial correlations through matrix multiplication and Softmax operations. (b) Local spatial attention module, which enhances fine-grained details using depth-wise convolutions (DWconv) and nonlinear activations (Swish, Tanh), adaptively adjusting feature weights for localized spatial regions.

the previous velocities on the current velocity, c_1 and c_2 are acceleration coefficients, which control the maximum step size the velocity of the particle can be changed due to personal and global best positions, respectively, r_1 and r_2 are random numbers in the range [0, 1], providing stochasticity to the search process, and p_i denotes the personal best position of particle i (personal best).

Based on the aforementioned optimization approach, the SADMM includes both local and global spatial attention modules, aimed at enhancing change detection processes by capturing global contextual relationships across spatial dimensions. Similarly, the SADMM employs principles akin to those of PSO, namely, using both local and global optimum information to dynamically adjust processing strategies. In the SADMM, this adjustment is manifested in the optimization of attention weights, simulating the update mechanism of particle positions and velocities in PSO, allowing the module to dynamically allocate computing resources and attention across different regions of hyperspectral data, thus effectively pinpointing the most apparent areas of change in the image.

The global spatial attention module enhances image processing by focusing on the global attention of the feature map, capturing contextual relationships across spatial dimensions, thereby improving the capability of the model to process global features. This module uses a multihead attention mechanism, where input features are first expanded through a fully connected layer and then decomposed into queries, keys, and values to handle complex inputs without sacrificing the range of perception. By average pooling the keys and values, it effectively generalizes the broader spatial context. The attention mechanism computes scores at each position, which are then normalized by a Softmax function, weighting the pooled values to emphasize regions of the image with higher information significance.

Algorithm 2 Spatial Attention-Enhanced Difference Mapping for HSI Change Detection

Input: Initial feature map $\mathbf{D}^{(0)}$ obtained from $\mathbf{X}_1$ and $\mathbf{X}_2$;
Output: The enhanced feature map $\mathbf{D}^{(k)}$;
Set w, c_1, c_2, r_1, $r_2 > 0$, $k = 0$;
Initialize update rate $\mathbf{V}^{(0)} = 0$;
while not converged **do**
Update $\mathbf{D}^{(k+1)}$ via(26);
Update $\mathbf{V}^{(k+1)}$ via(27);
$k = k + 1$;
end while
return The enhanced feature map $\mathbf{D}$.

The local spatial attention module emphasizes enhancing the discriminative ability for local features through a multihead attention mechanism coupled with depthwise separable convolutions. This module initiates its process by adapting the input features for different heads of the attention mechanism, where each head independently focuses on various aspects of the spatial details. Subsequently, depthwise separable convolutions refine these features by processing them independently across channels, enhancing the capture of local patterns effectively. Element-wise multiplication of the processed queries and keys highlights areas with high feature correlation, while subsequent linear transformations through Swish and Tanh activation functions further enrich the representation capabilities. This attention module significantly sharpens the focus of the model, focusing on important local features, thus improving the accuracy of hyperspectral change detection tasks. Specific details of the iterative process of the two modules can be seen in Fig. 4. This iterative process can be expressed as

$$\mathbf{D}^{(k+1)} = \mathbf{D}^{(k)} + \mathbf{V}^{(k)} \tag{26}$$

$$\mathbf{V}^{(k+1)} = w\mathbf{V}^{(k)} + c_1 r_1 \left(L\left(\mathbf{D}^{(k+1)}\right) - \mathbf{D}^{(k+1)}\right) + c_2 r_2 \left(G\left(\mathbf{D}^{(k+1)}\right) - \mathbf{D}^{(k+1)}\right) \tag{27}$$

where w, c_1, c_2, r_1, and $r_2 > 0$ are learnable parameters that are continuously fine-tuned during training, $\mathbf{D}^{(k)}$ denotes the feature map obtained at the kth iteration, $\mathbf{V}^{(k)}$ denotes the rate used to update $\mathbf{D}$ at the kth iteration, L is the local spatial attention module, and G is the global spatial attention module. The algorithm flow of the SADMM is described in Algorithm 2.

F. Loss Function

In hyperspectral change detection tasks, cross-entropy loss is a commonly used method to measure the difference between the model output and the actual label. Specifically, cross-entropy loss evaluates the accuracy of the prediction of the model for each pixel change in state. The goal is to minimize the distance between the predicted probability distribution of the model output and the true probability distribution of the

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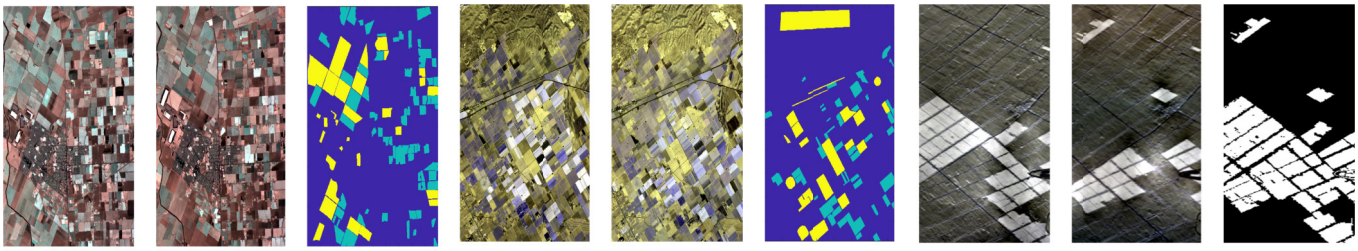


Fig. 5. Bay Area, Santa Barbara, and China datasets. (a) and (b) HSIs from the Bay Area dataset acquired in 2013 and 2015, respectively. (c) Ground-truth map for the Bay Area dataset. (d) and (e) HSIs from the Santa Barbara dataset acquired in 2013 and 2014, respectively. (f) Ground-truth map for the Santa Barbara dataset. (g) and (h) HSIs from the China dataset acquired on May 3, 2006, and April 23, 2007, respectively. (i) Ground-truth map for the China dataset.

target. The cross-entropy loss is defined as follows:

$$L_{\mathrm{CE}} = -\frac{1}{N}\sum_{n=1}^{N}\left[y_n \log \hat{y}_n + (1-y_n)\log\left(1-\hat{y}_n\right)\right] \tag{28}$$

where y_n represents the true label of the nth sample, $\hat{y}_n$ is the predicted probability for the nth sample, and N is the total number of samples. At the same time, using the mean square error (mse) to define the reconstruction loss is an effective method. This loss function is used to quantify the difference between the image reconstructed by the model through the estimated abundance matrix and the original image. The mse loss can be expressed as follows:

$$L_{\mathrm{REC}} = \frac{1}{N}\sum_{n=1}^{N}\left(\left\|\mathbf{X}_{1,n}-\hat{\mathbf{X}}_{1,n}\right\|_F^2 + \left\|\mathbf{X}_{2,n}-\hat{\mathbf{X}}_{2,n}\right\|_F^2\right) \tag{29}$$

where $\mathbf{X}_{i,n}$ represents the original image at time point i for the nth sample and $\hat{\mathbf{X}}_{i,n} = \mathbf{A}\mathbf{S}_{i,n}$ represents the reconstructed image, obtained by multiplying the shared endmember matrix $\mathbf{A}$ with the abundance matrix $\mathbf{S}_{i,n}$ corresponding to the respective time point. In this article, we use multiple convolutional layers to represent $\mathbf{A}$, which serves as a learnable approximation of the spectral mixing operator and is optimized in an end-to-end manner without explicit endmember estimation. $\|\cdot\|_F$ denotes the Frobenius norm, which calculates the sum of the squared differences of all elements in the matrices. The total loss function is the sum of these two losses and can be expressed as follows:

$$L = L_{\mathrm{CE}} + \eta L_{\mathrm{REC}} \tag{30}$$

where η is a scaling factor that adjusts the contribution of the reconstruction loss to ensure it is appropriately weighted relative to the cross-entropy loss.

IV. Experimental Results and Analysis

A. Data Description and Experimental Setup

The effectiveness of hyperspectral change detection methods is evaluated across three public datasets: Bay Area, Santa Barbara, and China (see Fig. 5). Each dataset includes two HSIs taken at different times, accompanied by ground-truth maps that highlight changed and unchanged regions. The Bay Area and Santa Barbara datasets were collected by the AVIRIS sensor in 2013–2015 and 2013–2014, respectively, covering urban, mountainous, and agricultural areas in California. The China dataset was acquired by the Hyperion sensor onboard the EO-1 satellite in 2006 and 2007, covering farmland in Yancheng, China. All datasets share a spectral range of 0.4–2.5 μm with a spectral resolution of 10 nm. AVIRIS provides 224 bands at 20-m spatial resolution, while Hyperion offers 154 bands at 30-m spatial resolution. Detailed information is provided in Table I.

TABLE I
Detailed Information on the Three Datasets

Dataset	Bay Area	Santa Barbara	China
Date T1	2013	2013	May 3, 2006
Date T2	2015	2014	Apr. 23, 2007
Image Size	600 × 500	984 × 740	420 × 140
Spectral Bands	224	224	154
Unchanged Pixels	34,211	80,418	40,417
Changed Pixels	39,270	52,134	18,383

B. Evaluation Metrics

In this study, we use several metrics to quantitatively evaluate the performance of our model, including overall accuracy (OA), kappa coefficient (KC), overall error (OE), and the F1-score. OA measures the proportion of correct classifications, while KC evaluates the consistency between model predictions and actual observations, accounting for random chance. OE represents the total number of misclassified samples, indicating misclassification frequency. The F1-score combines precision and recall, assessing the model's ability to balance both false positives and false negatives. These metrics collectively offer a comprehensive analysis of the model's detection performance.

C. Benchmark Methods and Experimental Protocols

To validate the proposed LSU-SADM method, we compare it with several widely used change detection methods, including unsupervised approaches like CVA [8], IR-MAD [11], and PCA [9], supervised methods like the SVM [13], and deep learning models such as the ReCNN [49], ML-EDAN [22], AMCAN-HMSchA [35], and MSDFFN [50]. CVA analyzes spectral differences to identify changes, while IR-MAD

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TABLE II
QUANTITATIVE ASSESSMENT OF LSU-SADM AND OTHER STATE-OF-THE-ART CHANGE DETECTION METHODS FOR THREE DATASETS

Dataset	Bay Area				Santa Barbara				China			
Methods	OA	KC	OE	F1-score	OA	KC	OE	F1-score	OA	KC	OE	F1-score
CVA	0.8406	0.6844	11710	0.8344	0.8325	0.6519	22204	0.7920	0.8749	0.6998	7357	0.7882
IR-MAD	0.8905	0.7826	8046	0.8883	0.8739	0.7264	16714	0.8223	0.8974	0.7632	6034	0.8382
PCA	0.8433	0.6897	11515	0.8368	0.8088	0.5890	25342	0.7381	0.8892	0.7322	6517	0.8099
SVM	0.9226	0.8446	5687	0.9272	0.9187	0.8312	10783	0.8993	0.9005	0.7724	5852	0.8458
ReCNN	0.9341	0.8680	4841	0.9373	0.9426	0.8787	7612	0.9253	0.9292	0.8387	4165	0.8911
ML-EDAN	0.9707	0.9413	2151	0.9723	0.9724	0.9421	3655	0.9648	0.9404	0.8638	3506	0.9078
AMCAN-HMSchA	0.9650	0.9298	2573	0.9668	0.9651	0.9265	4627	0.9549	0.9275	0.8344	4263	0.8879
MSDFFN	0.9818	0.9633	1341	0.9831	0.9837	0.9657	2165	0.9791	0.9468	0.8782	3131	0.9174
LSU-SADM	**0.9842**	**0.9683**	**1158**	**0.9852**	**0.9905**	**0.9801**	**1256**	**0.9879**	**0.9494**	**0.8845**	**2977**	**0.9219**

improves MAD by iteratively adjusting pixel weights for better accuracy. PCA reduces dimensionality to enhance computational efficiency. The SVM classifies changes by creating a hyperplane. The ReCNN combines the RNN and CNN features for spectral and spatial extraction. ML-EDAN uses a multilevel encoder–decoder network to optimize accuracy. AMCAN-HMSchA focuses on subpixel correlation differences using hierarchical attention, improving sensitivity, and MSDFFN integrates multiscale features to enhance the detection of subtle changes.

This study conducts all its experiments using the PyTorch platform, running on an NVIDIA GTX 3060 GPU. For the training process, the Adam optimizer is employed with a learning rate initially set at 0.001, which is scheduled to be halved every 20 epochs. We have configured the batch size to 16, and the training is programmed to terminate after 200 epochs. Additionally, the patch size for input data is selected as 11×11 to capture adequate spatial context.

D. Experimental Results

The analysis of change detection capabilities across various methodologies, as demonstrated through visual results and quantitative metrics, highlights the distinct performances among traditional methods, modern deep learning approaches, and the advanced LSU-SADM method.

The quantitative results in Table II show that traditional methods, such as the CVA, IR-MAD, SVM, and PCA, perform well in simple cases but struggle in complex environments. For instance, in the Bay Area dataset, CVA has the highest OE of 11710, while IR-MAD follows at 8046, highlighting their challenges in detecting changes accurately. While the SVM and PCA reduce false positives and improve noise control, the SVM achieves an OE of 5687 in the Bay Area. However, these methods still occasionally miss subtle changes that are crucial for accurate landscape analysis.

Deep learning methods such as the ReCNN, ML-EDAN, and AMCAN-HMSchA mark a significant advancement. Utilizing complex neural network architectures, these methods extract detailed spatial and spectral features, significantly enhancing change detection. In the Santa Barbara dataset, the ReCNN achieves an F1-score of 0.9253, and ML-EDAN reaches 0.9648, demonstrating their superior capability to manage both broad and subtle changes. Visually, these methods produce cleaner and more accurate difference maps compared to traditional methods.

LSU-SADM stands out with its exceptional performance metrics across all datasets. In the China dataset, it not only achieves the highest F1-score of 0.9219 but also maintains the lowest OE of 2977, underscoring its precision in detecting relevant changes with minimal noise. The visual results corroborate this data, showing the difference maps of LSU-SADM to be markedly clearer and more coherent, with significantly reduced misclassifications compared to all other methods. The visualization results produced by various competing methods across three commonly used datasets are displayed in Figs. 6–8.

E. Ablation Study

1) Number of Endmembers: We conducted experiments with different numbers of endmembers while keeping other parameters fixed to evaluate the impact on change detection accuracy and the smoothness [51] of the abundance matrices. Abundance matrix smoothness refers to the spatial consistency of material abundances across neighboring pixels, with lower values indicating more gradual and realistic variations. This metric is crucial because smoother abundance maps reflect a more accurate physical distribution of materials, helping to reduce redundancy and improve feature representation.

For change detection accuracy (Fig. 9), 20 endmembers generally produced the best results across all datasets. In the China dataset, for instance, 20 endmembers achieved an OA of 0.9494, a Kappa of 0.8845, and the lowest OE of 2977. Reducing or increasing the number of endmembers slightly lowered OA and Kappa while increasing OE. Regarding smoothness (Fig. 10), 20 endmembers performed best for the Bay Area, while the Santa Barbara and China datasets showed slight improvements with fewer endmembers. However, the impact on accuracy remained minimal, with differences in

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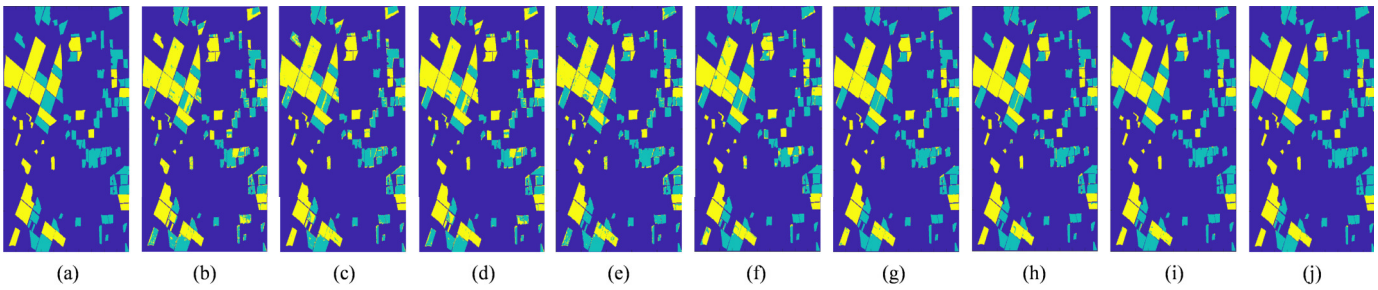


Fig. 6. Comparative visualization of methods for the Bay Area dataset. (a) Ground-truth map. (b) CVA. (c) IR-MAD. (d) PCA. (e) SVM. (f) ReCNN. (g) ML-EDAN. (h) AMCAN-HMSchA. (i) MSDFFN. (j) LSU-SADM.

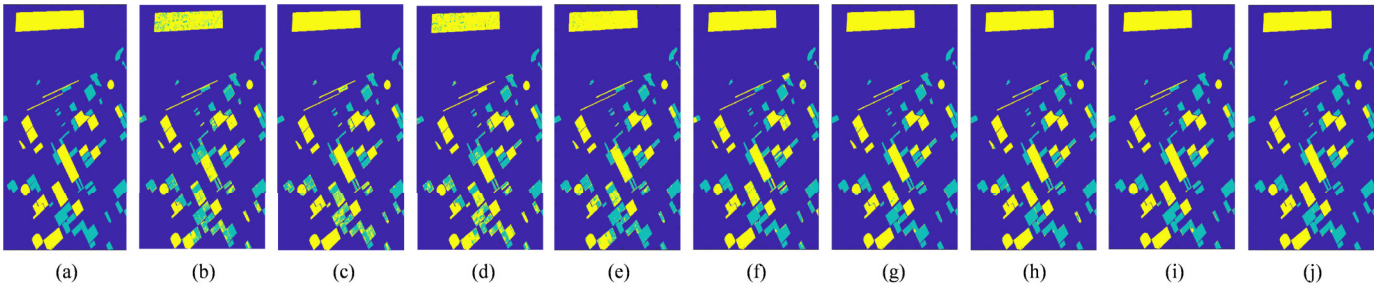


Fig. 7. Comparative visualization of methods for the Santa Barbara dataset. (a) Ground-truth map. (b) CVA. (c) IR-MAD. (d) PCA. (e) SVM. (f) ReCNN. (g) ML-EDAN. (h) AMCAN-HMSchA. (i) MSDFFN. (j) LSU-SADM.

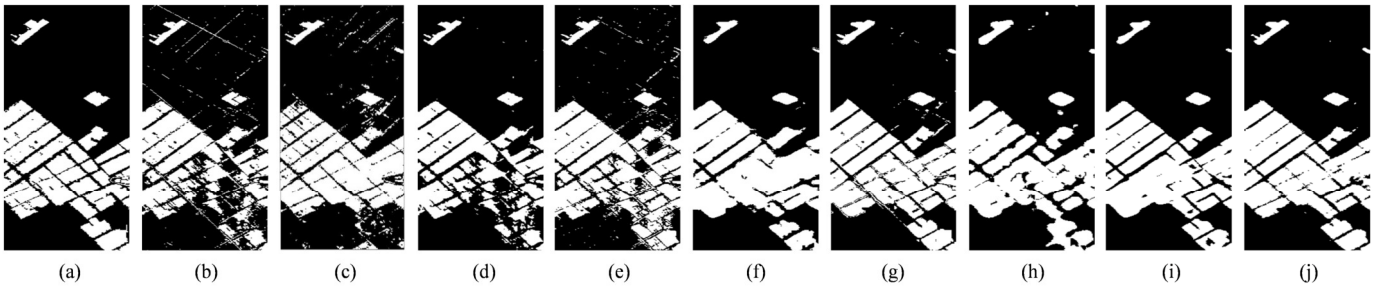


Fig. 8. Comparative visualization of methods for the China dataset. (a) Ground-truth map. (b) CVA. (c) IR-MAD. (d) PCA. (e) SVM. (f) ReCNN. (g) ML-EDAN. (h) AMCAN-HMSchA. (i) MSDFFN. (j) LSU-SADM.

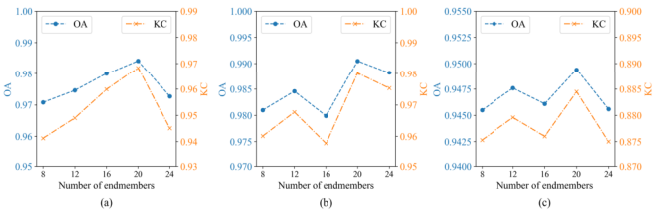


Fig. 9. Performance trends of the proposed method across three datasets with different numbers of endmembers. (a) Bay Area dataset. (b) Santa Barbara dataset. (c) China dataset.

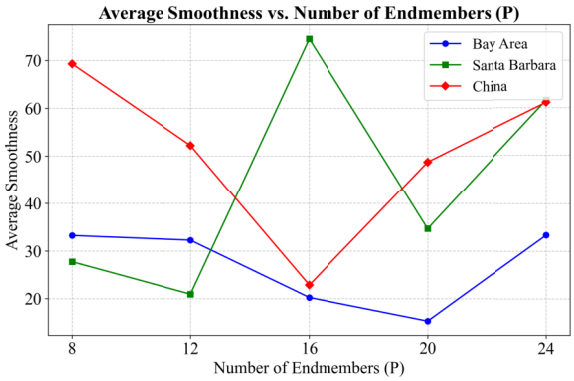


Fig. 10. Relationship between the number of endmembers (P) and the average smoothness of abundance maps for three datasets: Bay Area, Santa Barbara, and China.

China within 0.01%. Therefore, 20 endmembers were selected as optimal for all three datasets, balancing both accuracy and feature concentration. The abundance matrix visualizations are shown in Fig. 11.

2) Patch Size Selection for Change Detection: In our experiments on the Bay Area dataset, we assess the impact of patch size on change detection accuracy. As shown in Fig. 12, smaller patch sizes (5×5 and 7×7) maintain high OA but yield lower F1-scores of 0.9735 and 0.9675, indicating challenges in capturing sufficient spatial details. Increasing the patch size to 9×9 further reduces the F1-score to 0.9655 due to the

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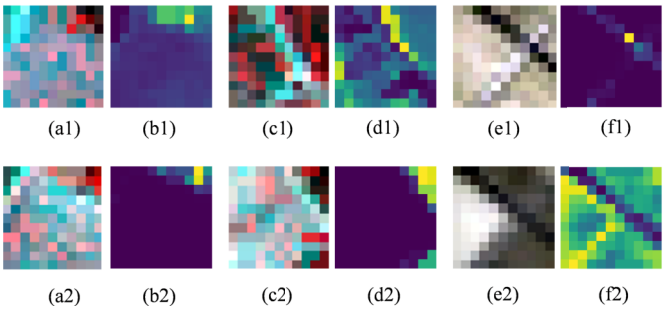


Fig. 11. Bi-temporal hyperspectral patches and their corresponding abundance maps from the Bay Area, Santa Barbara, and China datasets. Specifically, (a1), (a2), (c1), (c2), (e1), and (e2) input hyperspectral patches at two time points, while (b1), (b2), (d1), (d2), (f1), and (f2) correspond to abundance maps extracted by the proposed unmixing module.

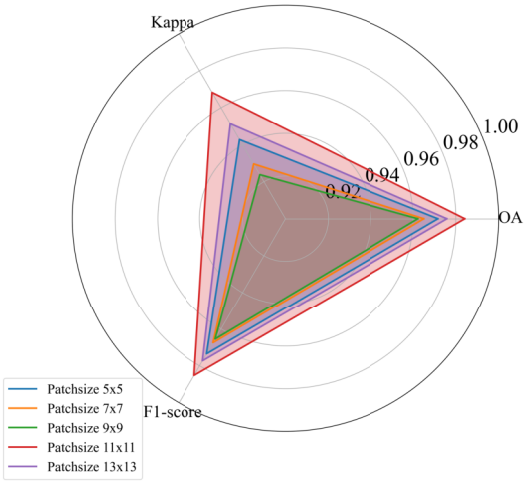


Fig. 12. Detection performance of different patch sizes.

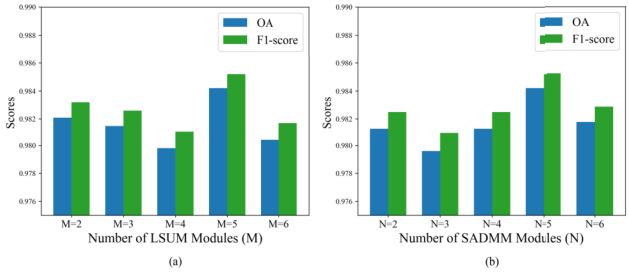


Fig. 13. Impact of module quantity on hyperspectral change detection accuracy. (a) Effect of LSUM quantity on performance. (b) Effect of SADMM quantity on performance.

dilution of spectral features. However, an 11×11 patch size strikes the optimal balance, improving the F1-score to 0.9852. Larger patches, such as 13×13, maintain good performance (F1-score of 0.9773) but show slight declines, likely due to excessive averaging. Overall, the 11×11 patch size is optimal for maximizing detection accuracy by effectively integrating spatial and spectral information.

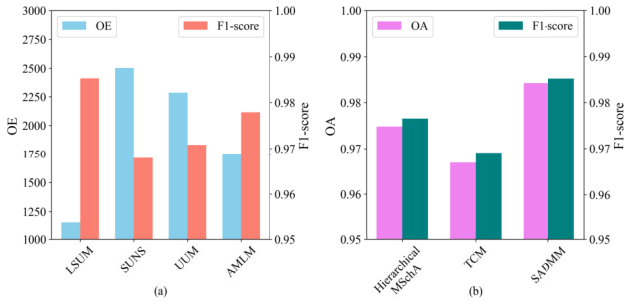


Fig. 14. Performance comparison of modules in hyperspectral change detection. (a) Comparison across unmixing modules. (b) Comparison across attention modules.

3) Impact of LSUM Quantity on Change Detection Accuracy: In our ablation study on the Bay Area dataset, we assess how varying the number of hyperspectral LSUMs impacts change detection accuracy while keeping the SADMMs fixed at five. As shown in Fig. 13(a), five LSUM achieves the best performance with an OA of 0.9842, a KC of 0.9683, and an F1-score of 0.9852. Reducing the LSUM count to four leads to a drop in OA (0.9798) and F1-score (0.9811), indicating decreased unmixing effectiveness. Increasing the count to six results in a slight performance decline, suggesting potential overfitting or increased computational load. This confirms that both too few and too many LSUM modules can hinder model performance, with five LSUM providing the optimal balance.

4) Impact of SADMM Quantity on Change Detection Accuracy: The number of SADMMs plays a key role in hyperspectral change detection accuracy. We conducted experiments on the Bay Area dataset with a fixed number of 5 hyperspectral LSUMs, assessing the impact of different SADMM quantities. As shown in Fig. 13(b), using five SADMM modules yields the highest OA of 0.9842, indicating optimal use of spatial attention mechanisms. Reducing SADMM to four or three decreases OA to 0.9813 and 0.9796, respectively, due to reduced spatial resolution. Increasing SADMM to six results in a slight OA improvement to 0.9817, suggesting marginal gains at the cost of added complexity. This highlights that an appropriate number of SADMM modules is essential for effective change detection.

5) Effectiveness of the LSUM: In a comparative study on the Bay Area dataset, the LSUM outperforms other unmixing modules, achieving an OE of 1158 and an F1-score of 0.9852. Replacing the LSUM with the Sparse Unmixing Network Structure [52] increases the OE to 2503 and reduces the F1-score to 0.9678. The United Unmixing Module [34] shows an OE of 2285 and an F1-score of 0.9708, while the Endmember Sharing-Based AMLM [35] achieves an OE of 1741 and an F1-score of 0.9779. Fig. 14(a) presents a bar graph of the results, emphasizing LSUM's superior performance in minimizing errors and maximizing the F1-score. This demonstrates LSUM's strength in handling complex spectral changes, with its integration of low-rank and sparse techniques optimizing spectral and spatial data processing.

6) Effectiveness of the SADMM: In the ablation study of the SADMM, we compare its performance with hierarchical

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MSchA-based DIEM [35] and the temporal correlation module (TCM) [34]. The SADMM outperforms both, achieving an OA of 0.9842 and an F1-score of 0.9852 on the Bay Area dataset, significantly surpassing hierarchical MSchA-based DIEM (OA: 0.9748, F1: 0.9765) and TCM (OA: 0.9668, F1: 0.9691). SADMM's superior performance (OE = 1158) in detecting changes and reducing misclassifications highlights its strength in enhancing spatial attention, especially in complex scenarios. Fig. 14(b) shows the experimental results, indicating that SADMM's direct focus on spatial details makes it more effective than the other methods in complex change detection tasks.

V. Conclusion

In this article, we introduce the interpretable LSU-SADM for hyperspectral change detection. This framework combines low-rank sparse unmixing techniques with spatial attention mechanisms, aiming to enhance accuracy and interpretability, particularly suitable for environmental monitoring and urban planning. LSU-SADM demonstrates significant performance improvements over traditional methods, efficiently extracting key material components and dynamically focusing on critical areas through its spatial attention module, thus precisely detecting subtle changes. This method not only delivers significant results but also offers high interpretability, making it valuable in applications where understanding the detection process is as crucial as the accuracy of the results themselves. Future research could explore the adaptability of this framework in various environments and attempt to integrate it with other deep learning architectures to boost feature extraction capabilities.

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